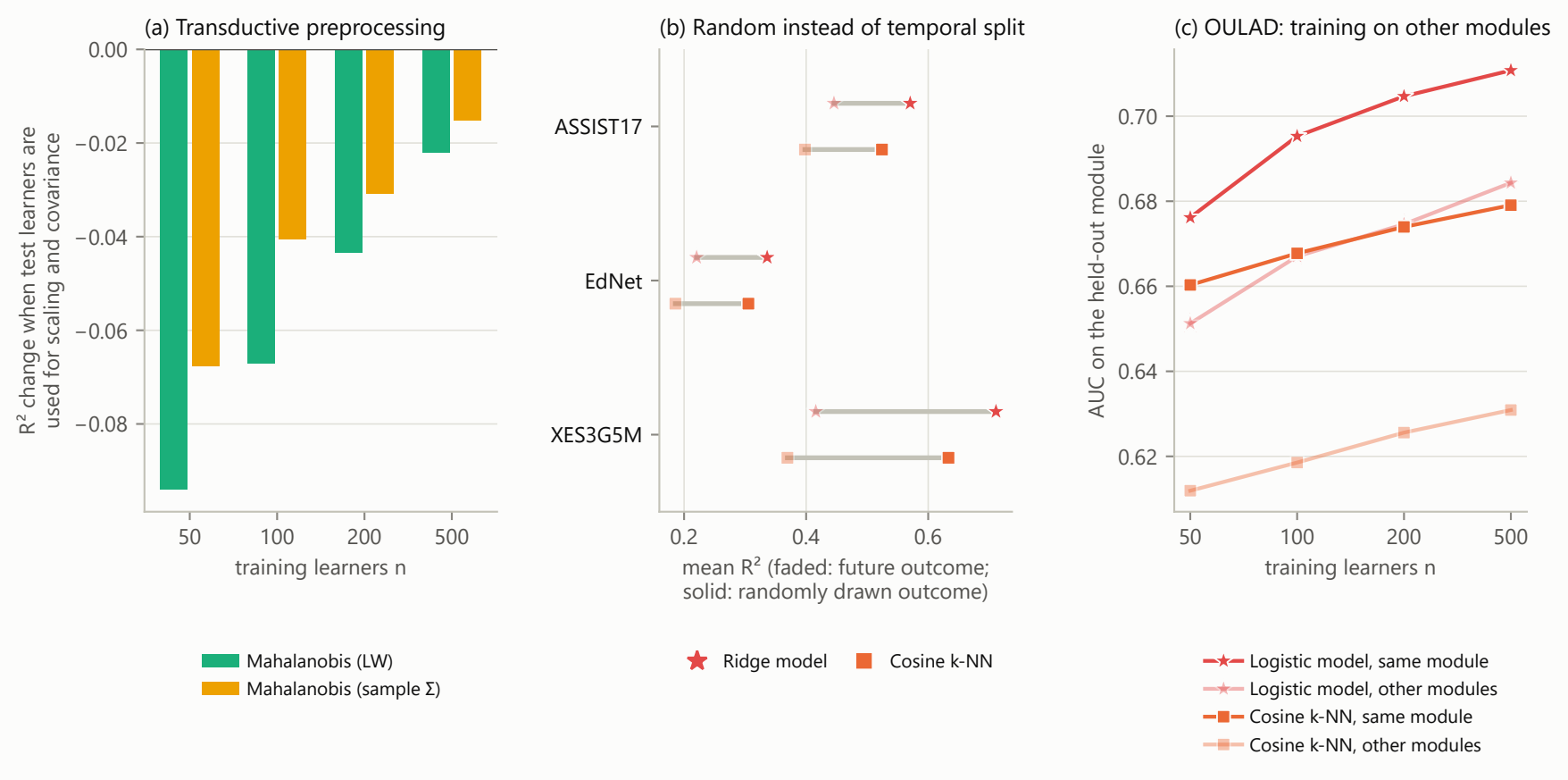
(a) Transductive preprocessing
0.00
−0.02
−0.04
−0.06
−0.08
R² change when test learners are used for scaling and covariance
50
100
200
500
training learners n
Mahalanobis (LW)
Mahalanobis (sample Σ)
(b) Random instead of temporal split
ASSIST17
EdNet
XES3G5M
0.2
0.4
0.6
mean R² (faded: future outcome; solid: randomly drawn outcome)
Ridge model
Cosine k-NN
(c) OULAD: training on other modules
0.70
0.68
0.66
0.64
0.62
AUC on the held-out module
50
100
200
500
training learners n
Logistic model, same module
Logistic model, other modules
Cosine k-NN, same module
Cosine k-NN, other modules